

P3

Heat & Temperature

Physics | Classes 6, 7 & 8 Combined

Classes 6 • 7 • 8 | TET Paper II Science

A • HEAT — Definition, Sources & Flow

Heat	Total Kinetic Energy of all constituent particles of an object — form of energy, not matter
SI Unit	Joule (J) also measured in calorie 1 calorie = 4.186 J
Temperature	Average Kinetic Energy of molecules — measures warmness or coldness of a substance
SI Unit (Temp)	Kelvin (K) also °C (Celsius) and °F (Fahrenheit)
Flow of Heat	Always flows from HIGHER temperature → LOWER temperature (like water from high to low level)
Thermal Contact	Two bodies that can exchange heat energy with each other
Thermal Equilibrium	Two bodies in thermal contact that NO longer affect each other's temperature — same temp reached

Sources of Heat (Class 6 — Section 1.1)

Sun	Primary source — gives heat besides light; barefoot walking on sunny afternoon is difficult
Combustion	Burning of fuels: wood, kerosene, coal, charcoal, petrol, LPG → produces heat energy
Friction	Rubbing two surfaces → heat generated (e.g., ancient humans rubbed stones to make fire)
Electricity	Current through conductor → heat (water heater, iron box, electric kettle work on this principle)

Heat vs Temperature — The Key Distinction (Class 6 — Section 1.5)

- Heat = TOTAL KE of all molecules | Temperature = AVERAGE KE of molecules
- Same temperature (100°C) — 5L boiling water has MORE HEAT than 1L boiling water (more molecules)
- Pond water has more HEAT than a hot cup of tea (despite tea's higher temperature) — much larger volume
- Heat depends on: (1) Temperature of substance (2) Number of molecules present (mass/volume)
- Sense of touch is NOT reliable to measure temperature — use a thermometer
- Neela's mistake: $80^{\circ}\text{C} + 80^{\circ}\text{C} \neq 160^{\circ}\text{C}$ — mixing equal volumes gives $\sim 80^{\circ}\text{C}$ (not additive)

■ Common Exam Traps — Heat & Temperature

- Heat and temperature are NOT the same — different units, different definitions.
- Heat flows from high temperature → low temperature, NOT from more heat → less heat.
- When two bodies reach thermal equilibrium, heat flow STOPS — both at same temperature.
- Adding two volumes of 80°C water does NOT give 160°C — temperature stays $\sim 80^{\circ}\text{C}$.

B • THERMOMETER — Types, Scales & Conversion

Thermometric Liquids — Mercury vs Alcohol

Property	Mercury (Hg)	Alcohol
Appearance	Opaque, shining silver	Transparent — coloured brightly for visibility
Boiling Point	357°C (high — wide range)	78°C (low — cannot measure high temperatures)
Freezing Point	-39°C	-100°C or lower (ideal for very cold temperatures)
Stickiness	Does NOT stick to glass walls	May slightly wet glass
Expansion	Uniform expansion (equal per °C)	Larger expansion per °C (more sensitive to change)
Heat conduction	Good conductor of heat	Poor conductor of heat
Suitable for	Wide range; clinical & lab use	Very low temperatures (Arctic, polar regions)

Types of Thermometers (Class 7 — Section 1.4.1 & 1.4.2)

Clinical Thermometer	Measures human body temperature range: 35°C to 42°C (94°F to 108°F)
Kink in Clinical Therm.	Prevents mercury from flowing back into bulb after removal from patient's mouth
Reading clinical therm.	Body temperature normally measured in °F (Fahrenheit more sensitive than Celsius)
Lab Thermometer	Measures higher temperatures for scientific research and industry — longer stem/bulb
Precaution	Never hold thermometer by bulb Jerk it down before use Do not heat in flame/sun

Temperature Scales — Celsius, Fahrenheit & Kelvin (Class 7 — Section 1.5)

Scale	Inventor / Named After	Freezing Pt of Water	Boiling Pt of Water	Notes
Celsius (°C)	Anders Celsius (Swedish astronomer, 1742); earlier called Centigrade (centium=100, gradus=step)	0°C	100°C	Centigrade = 100 steps; SI preferred unit for everyday use
Fahrenheit (°F)	Daniel Gabriel Fahrenheit (German physicist)	32°F	212°F	Used for body temperature; more sensitive scale (180 divisions vs 100 in Celsius)
Kelvin (K)	Lord Kelvin; SI unit of temperature — absolute scale	273.15 K	373.15 K	No negative values; starts from absolute zero (-273.15°C); used in science

°C to °F

$$(F - 32) / 9 = C / 5 \rightarrow F = (9/5 \times C) + 32 \text{ (or use: } F = 1.8C + 32)$$

°C to K

$$K = 273.15 + C \text{ (or } K = 273 + C \text{ approximately)}$$

Reference Point	Celsius (°C)	Fahrenheit (°F)	Kelvin (K)
Freezing point of water	0	32	273.15
Boiling point of water	100	212	373.15
Normal human body temperature	37	98.6	310.15
Average room temperature	23	72	296.15
Hottest natural temp (Earth)	56.7	134.06	329.85
Coldest natural temp (Earth)	-89	-128.2	184.15
Absolute zero	-273.15	-459.67	0

■ Did You Know?

- ◆ Celsius invented thermometer calibration in 1742 — original scale was reversed (100=freeze, 0=boil); later flipped by Linnaeus.
- ◆ Fahrenheit scale was first published in 1724 — based on brine (salt+ice) as 0°F and human body as 96°F originally.
- ◆ Lord Kelvin (William Thomson) proposed absolute temperature scale in 1848 — 0 K = absolute zero (no thermal motion).
- ◆ Hottest temperature ever recorded on Earth: 56.7°C in Libya, Africa (1922). Coldest: -89°C in Antarctica.

C · THERMAL EXPANSION — Solids, Liquids & Gases (Class 6 Sec. 1.7–1.9, Class 8 Sec. 4.1.1)

■ Thermal Expansion: Most substances EXPAND on heating and CONTRACT on cooling — same mass, more volume.

■ Molecules gain heat energy → move faster → spread apart → substance expands

■ Expansion maximum in GASES > LIQUIDS > SOLIDS (order of expansion)

Linear Expansion	Increase in LENGTH of a solid when heated
Cubical Expansion	Increase in VOLUME (in all 3 directions) of a solid when heated
Areal Expansion	Increase in AREA of a solid when heated (2D expansion)

■ Heated metal ball does not pass through ring → cubical expansion (Class 6 Activity 6 / Class 8 Activity 1)

■ Cycle spoke heating → length increases → bulb lights (circuit completed) → linear expansion proof

■ Electric wires sag in summer (expand), become taut in winter (contract) → left slack intentionally

Applications of Thermal Expansion (Class 6 — Section 1.9)

Railway track gaps	Small gaps between rail sections prevent buckling when rails expand in summer heat
Concrete bridge joints	Expansion joints allow bridge to expand/contract without cracking
Iron rim on wheel	Iron rim heated → expands → slipped onto wooden wheel → cools & contracts → holds tightly
Riveting	Hot rivet driven through hole in metal plates → head formed → cools & contracts → grips plates tightly
Glass cracking	Thick glass tumbler cracks — inner surface expands with hot liquid but outer surface stays cool
Borosilicate (Pyrex)	Very low thermal expansion → does not crack → used in kitchenware and laboratory glassware
Loose electric wires	Wires left slack intentionally so they can contract in winter without snapping

■ Expansion Order — Critical TET Fact

■ Expansion: Gas > Liquid > Solid (gases expand most for same temperature change)

■ All three types occur in solids: Linear (length), Areal (area), Cubical (volume)

■ In liquids and gases: only volume (cubical) expansion is significant

Process	Change of State	Heat	Example
Melting (Fusion)	Solid → Liquid	Absorbed ↑	Ice → Water at 0°C
Freezing (Solidification)	Liquid → Solid	Released ↓	Water → Ice at 0°C

Process	Change of State	Heat	Example
Vaporisation (Evaporation/Boiling)	Liquid → Gas	Absorbed ↑	Water → Steam at 100°C
Condensation	Gas → Liquid	Released ↓	Steam → Water; dew on cold surface
Sublimation	Solid → Gas (directly)	Absorbed ↑	Camphor, Dry ice (CO ₂), Naphthalene
Deposition	Gas → Solid (directly)	Released ↓	Frost formation on window pane

- Melting point of ice = 0°C | Boiling point of water = 100°C (at standard atmospheric pressure)
- Dry ice = solid CO₂ (carbon dioxide) sublimates at -57°C — used to refrigerate fish/meat in transit
- Water is the ONLY substance naturally found on Earth in all three states (solid, liquid, gas)
- Change of state is a PHYSICAL change — composition does not change, only state changes
- During change of state, temperature remains CONSTANT even though heat is being added/removed

Method	Medium	Mechanism	Daily Life Examples
Conduction	Solids (metals best) — also works between two touching solids	Heat passes from hot end to cold end by vibration of atoms — NO actual movement of atoms	Cooking in metal vessels; iron box heating cloth; wooden/plastic handles on utensils; igloo warm (snow = insulator)
Convection	Liquids and Gases (fluids) — NOT in solids	Hot fluid rises (less dense), cold fluid sinks — ACTUAL movement of molecules creates currents	Land breeze & sea breeze; hot air balloon rises; refrigerator cooling; room heater warms air; heating water in vessel
Radiation	No medium needed — works even through VACUUM	Energy travels as electromagnetic waves at speed of light — does not need matter	Sunlight warming Earth; feeling warmth of fire; black cooking vessel base absorbs more heat; white clothes in summer

Conductors vs Insulators (Conduction)

Good Conductors of Heat:

- All metals — copper, aluminium, iron, silver
- Used for: cooking vessels, iron box soleplate, heat exchangers
- Silver = best conductor of heat among metals
- Faster heat transfer → cooking quicker

Poor Conductors (Insulators):

- Wood, cork, cotton, wool, glass, rubber, plastic
- Used for: handles of utensils, oven mitts, building insulation
- Snow = poor conductor → inside igloo stays warm despite Arctic cold
- Air (still) = excellent insulator → woollen clothes trap air → warm

Land Breeze & Sea Breeze — Convection in Nature

	Land Breeze	Sea Breeze
When it occurs	Night	Day
Why	Land cools faster than sea at night → sea air warmer → rises → land air blows to sea	Land heats faster than sea during day → land air rises → cool sea breeze blows inland
Wind direction	Land → Sea (outward from coast)	Sea → Land (inward toward coast)

Radiation — Black & White Surfaces

Black surface	Absorbs MORE heat radiation — bottom of cooking vessels painted black for faster heating
White surface	Reflects heat radiation — white clothes in summer keep body cool
Visible radiation	When substance heated to $\sim 500^{\circ}\text{C}$, radiation becomes visible as dull red glow
Good absorber	Also a good emitter of radiation (black body radiates more than white body)

- Heat energy from Sun reaches Earth by RADIATION — travels 150 million km through vacuum

Units of Heat (Class 8 — Section 4.3.2)

Joule (J)	SI unit of heat energy heat is a form of energy → measured in joules
Calorie (cal)	Amount of heat to raise 1 gram of water by 1°C 1 calorie = 4.186 J
Kilocalorie	1 kcal = 4200 J (approximately) food energy is measured in kilocalories

Heat Capacity vs Specific Heat Capacity (Class 8 — Sections 4.3.3 & 4.3.4)

	Heat Capacity (C')	Specific Heat Capacity (C)
Definition	Amount of heat required to raise temperature of a GIVEN substance by 1°C or 1K	Amount of heat required to raise temperature of 1 kg of substance by 1°C or 1K
Symbol	C' (C prime)	C
Formula	$C' = Q / \Delta T$	$C = Q / (m \times \Delta T) \rightarrow Q = m \times C \times \Delta T$
SI Unit	J K^{-1} (Joule per Kelvin)	$\text{J kg}^{-1} \text{K}^{-1}$ (Joule per kilogram per Kelvin)
CGS Unit	$\text{cal}^{\circ}\text{C}$	$\text{cal g}^{-1} ^{\circ}\text{C}^{-1}$

- Specific Heat of Water = $4200 \text{ J kg}^{-1} \text{K}^{-1}$ | Water has the HIGHEST specific heat among common substances.**

- High specific heat of water → used as coolant in radiators, nuclear reactors, and engine cooling systems
- 100 g of water carries away MORE heat than 100 g of oil (water has higher heat capacity)
- Cooking oil heats up FASTER than water (lower specific heat capacity than water)
- 3 factors that affect heat gained/lost: (1) Mass of substance (2) Temperature change (3) Nature of material
- Specific heat capacity is a material property — different for every substance

Device	What it Does	Key Features	Inventor / Year
Calorimeter	Measures amount of heat gained or lost by a substance — used in calorimetry	Metal vessel (copper/aluminium) in insulating jacket + thermometer + stirrer + heating element	First ice-calorimeter: Antoine Lavoisier & Pierre-Simon Laplace (1782)
Thermostat	Maintains temperature of a place or device at a constant set value	Senses temperature → switches appliance ON or OFF when set temp reached; "thermo"=heat + "static"=same	Used in: AC, refrigerator, oven, water heater, iron box, room heater
Thermos Flask (Vacuum Flask)	Keeps contents hotter or cooler than surroundings for extended period	Double walls with vacuum between them; silvered inner walls reflect radiated heat back; conduction only at neck	Sir James Dewar (Scottish scientist, 1892) — also called Dewar Flask / Dewar Bottle

How Thermos Flask Prevents All Three Modes of Heat Transfer

Conduction prevented	Vacuum between double walls — no atoms/molecules to conduct heat
Convection prevented	Vacuum — no liquid or gas present to form convection currents
Radiation prevented	Silvered inner walls REFLECT radiated heat back into liquid — heat stays inside
Minor conduction path	Only at the glass neck (top) and insulated support at bottom — unavoidable but minimal

H · Solved Numerical Problems

Problem 1	Convert 37°C (normal body temperature) to Fahrenheit and Kelvin.
Given	$C = 37^{\circ}\text{C}$
Solution	$F = (9/5 \times 37) + 32 = 66.6 + 32 = 98.6^{\circ}\text{F}$ $K = 273.15 + 37 = 310.15\text{ K}$
Answer	98.6°F 310.15 K

Problem 2	Convert 212°F (boiling point) to Celsius.
Given	$F = 212^{\circ}\text{F}$
Solution	$C = (F - 32) \times 5/9 = (212 - 32) \times 5/9 = 180 \times 5/9 = 100^{\circ}\text{C}$
Answer	100°C (boiling point of water confirmed)

Problem 3	A metal ball at 30°C receives 3000 J. Temperature rises to 40°C. Find heat capacity.
Given	$Q = 3000\text{ J}$; $T_1 = 30^{\circ}\text{C}$; $T_2 = 40^{\circ}\text{C}$; $\Delta T = 40 - 30 = 10^{\circ}\text{C} = 10\text{ K}$
Solution	$C' = Q / \Delta T = 3000 / 10 = 300\text{ J K}^{-1}$
Answer	Heat Capacity = 300 J K⁻¹

Problem 4	84000 J raises 2 kg of water from 60°C to 70°C. Find specific heat capacity.
Given	$Q = 84000\text{ J}$; $m = 2\text{ kg}$; $\Delta T = 70 - 60 = 10^{\circ}\text{C} = 10\text{ K}$
Solution	$C = Q / (m \times \Delta T) = 84000 / (2 \times 10) = 84000 / 20 = 4200\text{ J kg}^{-1}\text{K}^{-1}$
Answer	Specific Heat Capacity = 4200 J kg⁻¹K⁻¹

Problem 5	Heat capacity of a vessel is 500 J K ⁻¹ . Find heat needed to raise temperature by 20 K.
Given	$C' = 500\text{ J K}^{-1}$; $\Delta T = 20\text{ K}$
Solution	$Q = C' \times \Delta T = 500 \times 20 = 10,000\text{ J}$
Answer	Q = 10,000 J = 10 kJ

Problem 6	Specific heat of metal = 160 J kg ⁻¹ K ⁻¹ . Find heat for 500 g metal, 125°C to 325°C.
Given	$C = 160\text{ J kg}^{-1}\text{K}^{-1}$; $m = 500\text{ g} = 0.5\text{ kg}$; $\Delta T = 325 - 125 = 200\text{ K}$
Solution	$Q = C \times m \times \Delta T = 160 \times 0.5 \times 200 = 16,000\text{ J}$
Answer	Q = 16,000 J = 16 kJ

I · Key Facts — Rapid Revision (Numbers, Years & Names only)

Formulas, Values & Units

Quantity / Concept	Formula / Value	Unit / Note
Heat (energy)	Total KE of all molecules	Joule (J)
Temperature	Average KE of molecules	Kelvin (K); also °C, °F
1 calorie	= 4.186 J (heat to raise 1g water by 1°C)	CGS unit of heat
1 kcal (food energy)	= 4200 J (approximately)	Dietary calorie
°C to °F	$F = (9/5 \times C) + 32$ OR $C = (F - 32) \times 5/9$	Conversion formula
°C to K	$K = 273.15 + C$	Conversion formula
Body temp	37°C = 98.6°F = 310.15 K	Normal human
Boiling pt of water	100°C = 212°F = 373.15 K	At 1 atm pressure
Freezing pt of water	0°C = 32°F = 273.15 K	At 1 atm pressure
Heat Capacity (C')	$C' = Q / \Delta T$	J K ⁻¹
Specific Heat (C)	$C = Q / (m \times \Delta T) \rightarrow Q = m \times C \times \Delta T$	J kg ⁻¹ K ⁻¹
Specific heat of water	4200 J kg ⁻¹ K ⁻¹ (highest common substance)	J kg ⁻¹ K ⁻¹
Clinical therm. range	35°C to 42°C = 94°F to 108°F	Human body only
Expansion order	Gas > Liquid > Solid	Thermal expansion
Heat transfer order	Conduction (solid) Convection (liquid/gas) Radiation (vacuum)	Medium needed
Absolute zero	-273.15°C = -459.67°F = 0 K	No thermal motion
Hottest on Earth	56.7°C (Libya, Africa, 1922)	Natural record
Coldest on Earth	-89°C (Antarctica)	Natural record

Scientists, Discoveries & Years

Scientist	Contribution	Year / Class
Anders Celsius	Invented Celsius (Centigrade) temperature scale	1742
Daniel G. Fahrenheit	Invented Fahrenheit scale; first mercury thermometer	1724
Lord Kelvin (W. Thomson)	Proposed absolute (Kelvin) temperature scale; SI unit named after him	1848
Antoine Lavoisier	First ice-calorimeter (with Laplace); identified oxygen role in combustion	1782
Pierre-Simon Laplace	Co-invented ice-calorimeter with Lavoisier	1782
Sir James Dewar	Invented vacuum (thermos) flask — also called Dewar flask / Dewar bottle	1892

J · Glossary — Definitions

Heat — Total kinetic energy of all constituent particles of a body; form of energy; SI unit: Joule

Temperature — Measure of average kinetic energy of molecules; measures hotness/coldness; SI unit: Kelvin

Thermal Contact — State of two bodies that can exchange heat energy between them

Thermal Equilibrium — State when two bodies in thermal contact no longer transfer heat — same temperature

Calorie — Amount of heat to raise 1 gram of water by 1°C; 1 cal = 4.186 J	Thermometer — Instrument used to measure temperature accurately and quantitatively
Clinical Thermometer — Thermometer for human body temperature; range 35°C to 42°C; has a kink	Lab Thermometer — Thermometer for scientific/industrial use; wider temperature range than clinical
Celsius Scale (°C) — Temperature scale: 0°C = freezing point, 100°C = boiling point of water; named after Anders Celsius	Fahrenheit Scale (°F) — Temperature scale: 32°F = freezing, 212°F = boiling; named after Daniel G. Fahrenheit
Kelvin Scale (K) — SI absolute temperature scale; 0 K = absolute zero; named after Lord Kelvin	Absolute Zero — The lowest possible temperature: 0 K = -273.15°C; no thermal motion of molecules
Thermal Expansion — Expansion of a substance when heated — most substances expand on heating	Linear Expansion — Increase in length of a solid on heating
Cubical Expansion — Increase in volume (in all three dimensions) of a substance on heating	Melting (Fusion) — Change of state from solid to liquid; heat is absorbed; e.g., ice → water at 0°C
Freezing — Change of state from liquid to solid; heat is released; e.g., water → ice at 0°C	Vaporisation — Change of state from liquid to gas; heat is absorbed; e.g., water → steam at 100°C
Condensation — Change of state from gas to liquid; heat is released; e.g., steam → water	Sublimation — Direct change of state from solid to gas without passing through liquid; e.g., camphor, dry ice
Deposition — Direct change of state from gas to solid; reverse of sublimation; e.g., frost formation	Conduction — Heat transfer in solids from hot to cold region — NO actual movement of atoms/molecules
Convection — Heat transfer in liquids and gases by actual movement of molecules from hot to cold regions	Radiation — Heat transfer as electromagnetic waves — can travel through vacuum; e.g., solar energy
Calorimetry — Technique to measure the amount of heat involved in a physical or chemical process	Calorimeter — Device to measure heat capacity of substances; metal vessel in insulating jacket
Heat Capacity (C') — Heat needed to raise temperature of given substance by 1°C or 1K; unit: JK ⁻¹	Specific Heat Cap. (C) — Heat needed to raise temperature of 1 kg of substance by 1°C or 1K; unit: J kg ⁻¹ K ⁻¹
Thermostat — Device that maintains constant temperature by switching appliance ON/OFF at set temperature	Thermos Flask — Insulating vessel with vacuum between double walls; keeps contents hot or cold; invented by Dewar
Dry Ice — Solid CO ₂ (carbon dioxide) — sublimates directly at -57°C; used to refrigerate perishables	Borosilicate Glass — Glass with very low thermal expansion; does not crack when heated; used in lab/kitchen
Igloo — Snow house; warm inside because snow is a poor conductor of heat (insulator)	